

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

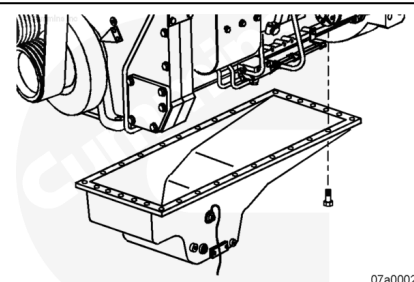
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7.
(</qs3/pubsys2/xml/en/procedures/57/57-007-037-tr.html>)

Remove

⚠ WARNING ⚠

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Remove the lubricating oil pan and gasket.

Discard the gasket.

Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear safety glasses or a face shield, as well as protective clothing, to reduce the possibility of personal injury when using a steam cleaner or high-pressure water.

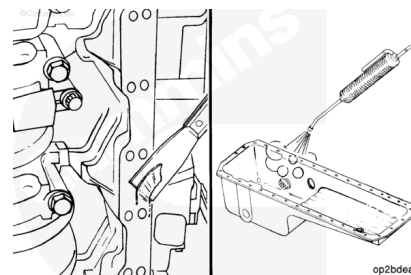
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Remove all gasket material from the cylinder block and oil pan surfaces.

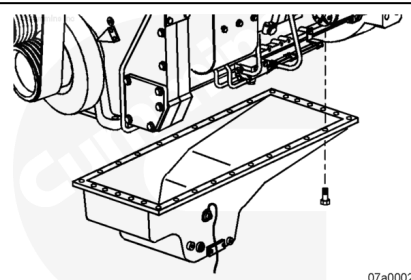
Steam-clean the lubricating oil pan.

Dry with compressed air.



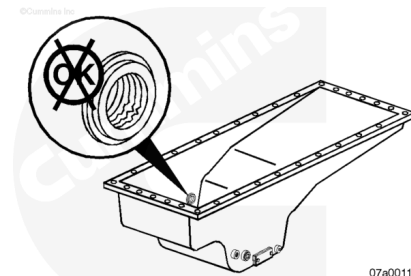
Note : If cracks are found, replace the damaged part. Do not attempt to repair the oil pan by welding.

Inspect the oil pan, suction tube, and mounting bracket for cracks and other damage.



Note : If the threaded holes can not be repaired with the taps listed in the table below, the oil pan must be replaced.

Inspect the threaded holes for damage. Damaged threads can be chased with the following taps.



Oil Pan Threaded Holes	
Thread Location	Tap Size
Temperature gauge	M14
Oil sump heater	M33
Oil drain	M27

Install

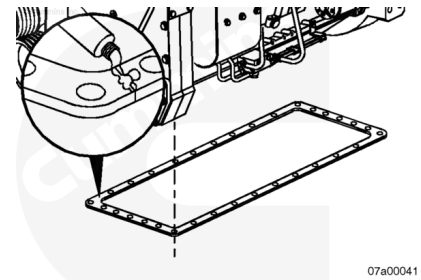
Verify the gasket surface with text present ("TS-9016" or "PAN SIDE") is facing toward the lubricating oil pan.

Note : The use of tie wraps will aid in holding the gasket to the block during the installation of the lubricating oil pan.

Check that the gaskets from the gear case gear cover and flywheel housing have been trimmed.

Apply RTV silicone adhesive/sealant, Part Number 3823494, to all gasket joints, and on the cylinder block, flywheel housing, gear housing and gear cover joints.

Install the gasket on the block.

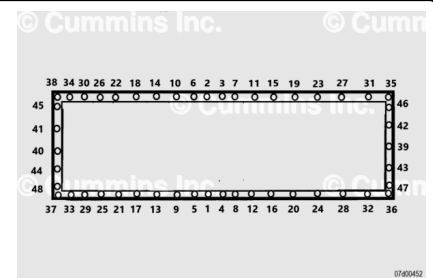


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Note : Some of the capscrew holes will be used to attach a bracket for the fuel lines.

Install the lubricating oil pan.

Tighten all capscrews in the sequence shown in the illustration.

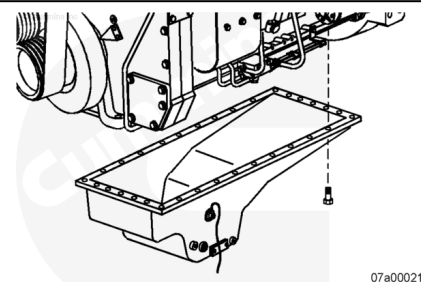


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Torque Value: 65 n•m [48 ft-lb]

Install the oil drain plug and tighten.

Torque Value: 47 n•m [35 ft-lb]



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Finishing Steps



Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil.

- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/57/57-007-037-tr.html>)

Last Modified: 16-Oct-2025
